

DONGYEONG KIM

Computer Science Undergraduate | AI Researcher
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RESEARCH INTERESTS

Neuromorphic computing, spiking neural networks, energy-efficient AI, computer vision, object detection, brain-inspired computing, and hardware-aware machine learning.

EDUCATION

Stony Brook University (SUNY Korea) | *B.S. in Computer Science* Feb. 2024 - Present

- Dean's List, Spring 2024.
- Selected coursework: Calculus I, Applied Linear Algebra, Discrete Mathematics, Foundations of Computer Science, and Programming Abstractions.

RESEARCH EXPERIENCE

Undergraduate Research Assistant | *Brain-Inspired Computing Lab (BCL)* Mar. 2024 - Present

- Conduct research on spiking neural networks, energy-efficient object detection, and neuromorphic computing under the direction of Prof. Yoon Seok Yang.
- Develop and evaluate Sigma-Delta spiking object-detection models, emphasizing accuracy, latency, energy, and deployment trade-offs.
- Implement training and evaluation workflows involving dataset preparation, temporal learning, surrogate-gradient methods, and computer-vision metrics.
- Conduct experiments with neuromorphic computing workflows, including Intel Lava and Intel Loihi 2 inference and performance evaluation.

PUBLICATIONS

Dongyeong Kim, Gilho Kim, Myeong Jun Kim, Si Yong Lee, and Yoon Seok Yang. "Advanced Training Algorithms in Sigma-Delta Spiking YOLO for Energy-Efficient Object Detection on Neuromorphic Hardware." *IEEE Access*, 2026. doi:10.1109/ACCESS.2026.3655178

Peer-reviewed open-access article. Developed and evaluated training algorithms for Sigma-Delta spiking neural networks for energy-efficient object detection and neuromorphic deployment.

ACADEMIC SERVICE AND EMPLOYMENT

Computer Science Department Senator | *SUNY Korea Student Government* 2024

- Represented Computer Science student interests in academic and student-governance discussions.

IT Services Administrative Assistant | *SUNY Korea Work-Study Program* Mar. - Jun. 2024

- Supported IT Services administrative and operational work for university users and internal processes.

TECHNICAL SKILLS

Programming	Python, Java, C / C++
ML and AI	PyTorch, deep learning, computer vision, object detection, efficient AI
Research	Spiking neural networks, neuromorphic computing, temporal learning, energy-aware evaluation
Tools	Linux, Jupyter, GitHub, Intel Lava, Intel Loihi 2